Lecture 23 (11/9)

Regularization

Today:

1) Questions about last week
2) Regularization: ridge regression
3) Regularization for the spike protein
4) Regularization for images

Practice questions

Question

Consider the covid spike protein example with three sites, where $X_j = 1$ if the mutation is present and $X_j = 0$ if not:

	X_1	X_2	X_3
mutation	Q498R	N501Y	Y505H

A variant is (X_1, X_2, X_3); for example $(1, 0, 1)$ has Q498R and Y505H. The model with all interactions is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + J_{12} X_1 X_2 + J_{13} X_1 X_3 + J_{23} X_2 X_3 + J_{123} X_1 X_2 X_3 + \epsilon$$

with

$$\beta_0 = 9.03, \quad \beta_1 = -0.25, \quad \beta_2 = 0.20, \quad \beta_3 = -0.90,$$
$$J_{12} = 0.55, \quad J_{13} = 0.45, \quad J_{23} = 0.68, \quad J_{123} = -0.43.$$

1) What is the expected difference in binding between the variants $(1, 1, 0)$ and $(0, 1, 1)$?
2) Does adding Q498R increase or decrease the expected binding? Compare the variants $(1, 0, 0)$ and $(0, 0, 0)$ with $(1, 1, 0)$ and $(0, 1, 0)$.

Practice questions

Question

A degree-K polynomial is fit to $N_{\text{train}} = 14$ training points. The training error is $\hat{\sigma}^2_{\text{train}} = 0.36$ and $\hat{\sigma}^2_Y = 1.80$ for the training Y values.

1) What is R^2 on the training points?
2) What are $\hat{\sigma}^2_{\text{train}}$ and R^2 when $K = 13$?

Practice questions

Question

A degree-K polynomial is fit to $N_{\text{train}} = 14$ training points. The training error is $\hat{\sigma}^2_{\text{train}} = 0.36$ and $\hat{\sigma}^2_Y = 1.80$ for the training Y values.

1) What is R^2 on the training points?
2) What are $\hat{\sigma}^2_{\text{train}}$ and R^2 when $K = 13$?

Question

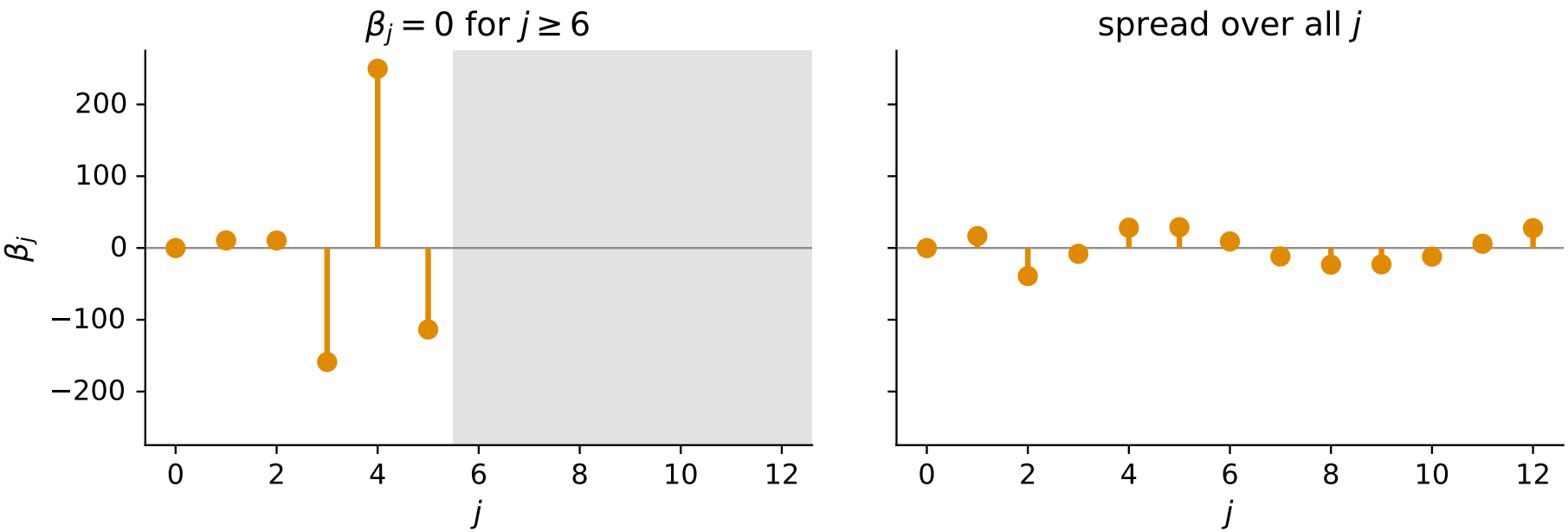
At $x = 0.56$ the degree-7 fit has bias 0.97, and the 95% confidence interval for $\hat{y}(0.56)$ has half-width 0.96. The noise has $\sigma_\epsilon = 0.5$.

1) Approximately what is $\text{var}_D\big(\hat{y}(0.56, D)\big)$?
2) What is the expected squared error $E\big[(\hat{y}(0.56, D) - Y)^2\big]$ for a new point at $x = 0.56$?

Questions about last week

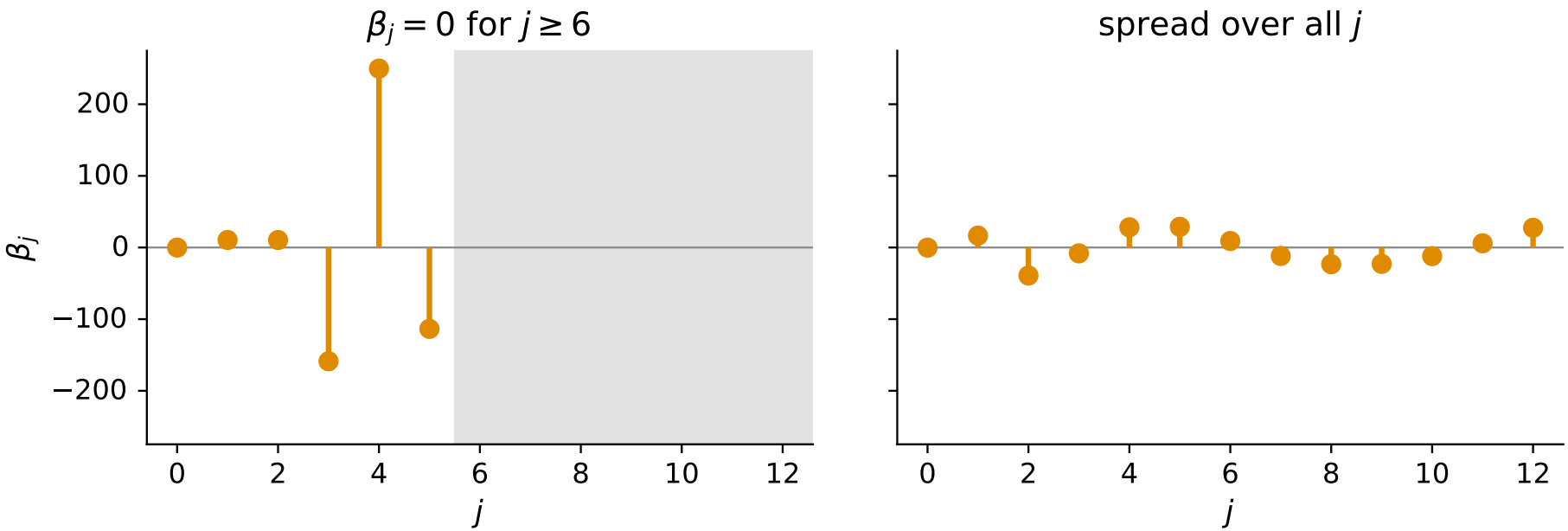
A hard cutoff, or coefficients that shrink?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots$$



A hard cutoff, or coefficients that shrink?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots$$



How do we implement the second case?

Example: estimating a mean

Start with the simplest model, $y = \beta_0$. Fitting it to data $Y_1, \ldots, Y_N$ means estimating the mean.

Example: estimating a mean

Start with the simplest model, $y = \beta_0$. Fitting it to data $Y_1, \ldots, Y_N$ means estimating the mean.

Least squares chooses β_0 to minimize the loss function

$$L(\beta_0) = \frac{1}{N} \sum_{i=1}^{N} (\beta_0 - Y_i)^2 = \hat{\sigma}^2_{\text{train}}.$$

This loss is just the training error, the average squared residual over the data used to fit the model.

Example: estimating a mean

Start with the simplest model, $y = \beta_0$. Fitting it to data $Y_1, \ldots, Y_N$ means estimating the mean.

Least squares chooses β_0 to minimize the loss function

$$L(\beta_0) = \frac{1}{N} \sum_{i=1}^{N} (\beta_0 - Y_i)^2 = \hat{\sigma}^2_{\text{train}}.$$

This loss is just the training error, the average squared residual over the data used to fit the model. A loss function is any function of the parameters that we minimize to fit a model.

Example: estimating a mean

Start with the simplest model, $y = \beta_0$. Fitting it to data $Y_1, \ldots, Y_N$ means estimating the mean.

Least squares chooses β_0 to minimize the loss function

$$L(\beta_0) = \frac{1}{N} \sum_{i=1}^{N} (\beta_0 - Y_i)^2 = \hat{\sigma}^2_{\text{train}}.$$

This loss is just the training error, the average squared residual over the data used to fit the model. A loss function is any function of the parameters that we minimize to fit a model.

$$\frac{dL}{d\beta_0} = \frac{2}{N} \sum_{i=1}^{N} (\beta_0 - Y_i) = 2(\beta_0 - \overline{Y}) = 0$$

Example: estimating a mean

Start with the simplest model, $y = \beta_0$. Fitting it to data $Y_1, \ldots, Y_N$ means estimating the mean.

Least squares chooses β_0 to minimize the loss function

$$L(\beta_0) = \frac{1}{N} \sum_{i=1}^{N} (\beta_0 - Y_i)^2 = \hat{\sigma}^2_{\text{train}}.$$

This loss is just the training error, the average squared residual over the data used to fit the model. A loss function is any function of the parameters that we minimize to fit a model.

$$\frac{dL}{d\beta_0} = \frac{2}{N} \sum_{i=1}^{N} (\beta_0 - Y_i) = 2(\beta_0 - \overline{Y}) = 0$$

$$\hat{\beta}_0 = \overline{Y}$$

Regularization

To pull β_0 toward 0, add a penalty to the loss:

$$L_\lambda(\beta_0) = \frac{1}{N}\sum_{i=1}^{N}(\beta_0 - Y_i)^2 + \lambda\beta_0^2$$

Regularization

To pull β_0 toward 0, add a penalty to the loss:

$$L_\lambda(\beta_0) = \frac{1}{N}\sum_{i=1}^{N}(\beta_0 - Y_i)^2 + \lambda\beta_0^2$$

$$\frac{dL_\lambda}{d\beta_0} = 2(\beta_0 - \overline{Y}) + 2\lambda\beta_0 = 0$$

Regularization

To pull β_0 toward 0, add a penalty to the loss:

$$L_\lambda(\beta_0) = \frac{1}{N}\sum_{i=1}^{N}(\beta_0 - Y_i)^2 + \lambda\beta_0^2$$

$$\frac{dL_\lambda}{d\beta_0} = 2(\beta_0 - \overline{Y}) + 2\lambda\beta_0 = 0$$

$$\hat{\beta}_0^\lambda = \frac{\overline{Y}}{1+\lambda}$$

Regularization

To pull β_0 toward 0, add a penalty to the loss:

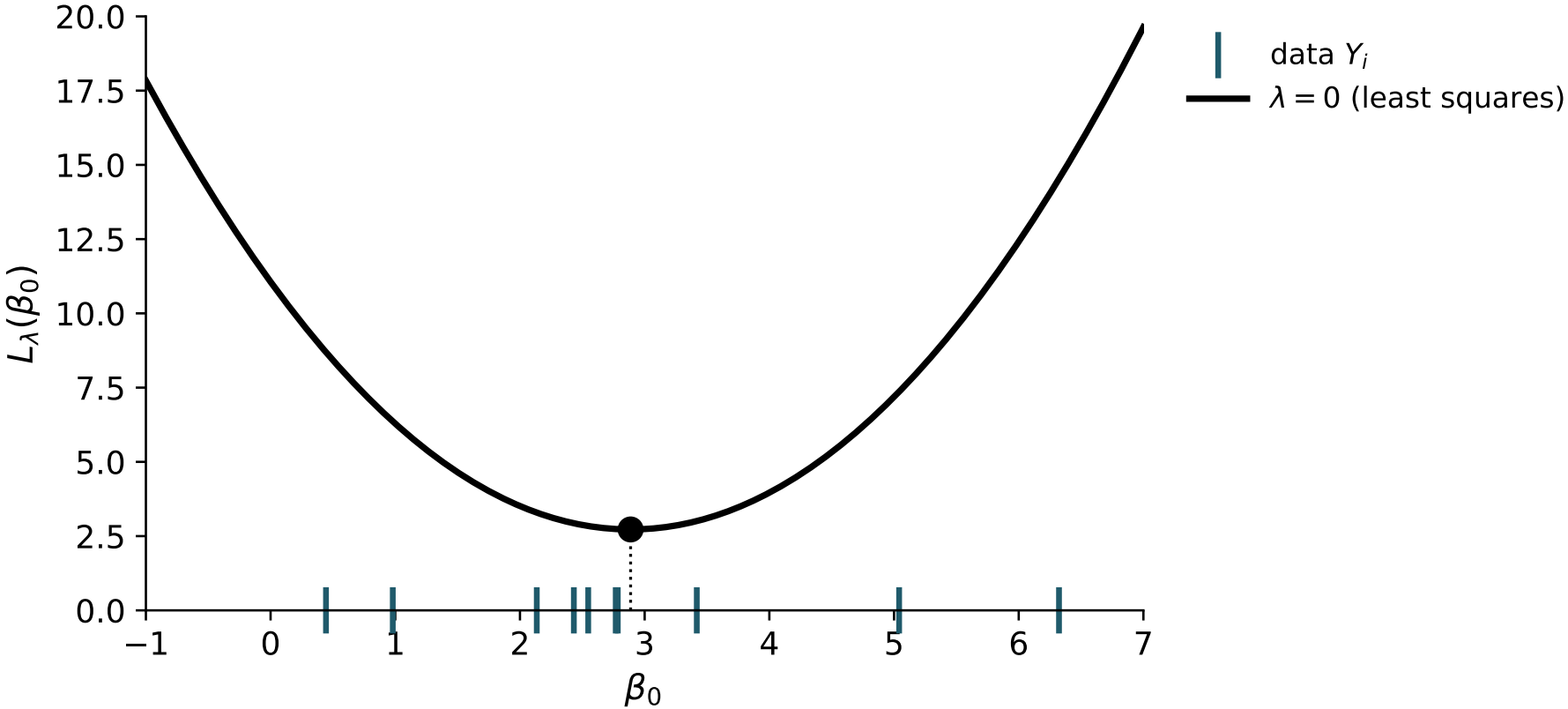
$$L_\lambda(\beta_0) = \frac{1}{N}\sum_{i=1}^{N}(\beta_0 - Y_i)^2 + \lambda\beta_0^2$$

$$\frac{dL_\lambda}{d\beta_0} = 2(\beta_0 - \overline{Y}) + 2\lambda\beta_0 = 0$$

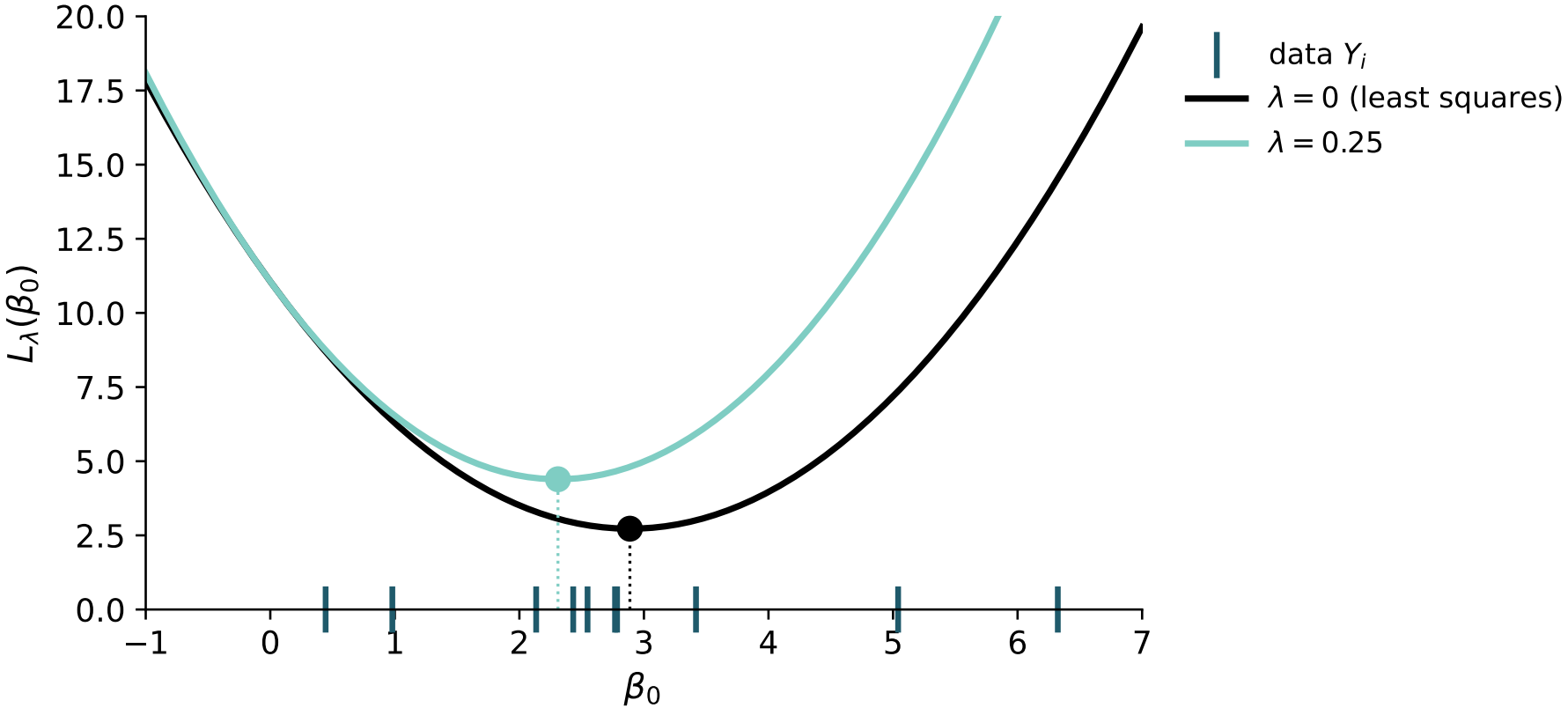
$$\hat{\beta}_0^\lambda = \frac{\overline{Y}}{1+\lambda}$$

$\lambda = 0$ gives $\overline{Y}$, and as $\lambda \to \infty$ the estimate goes to 0. The larger λ, the smaller $\hat{\beta}_0^\lambda$.

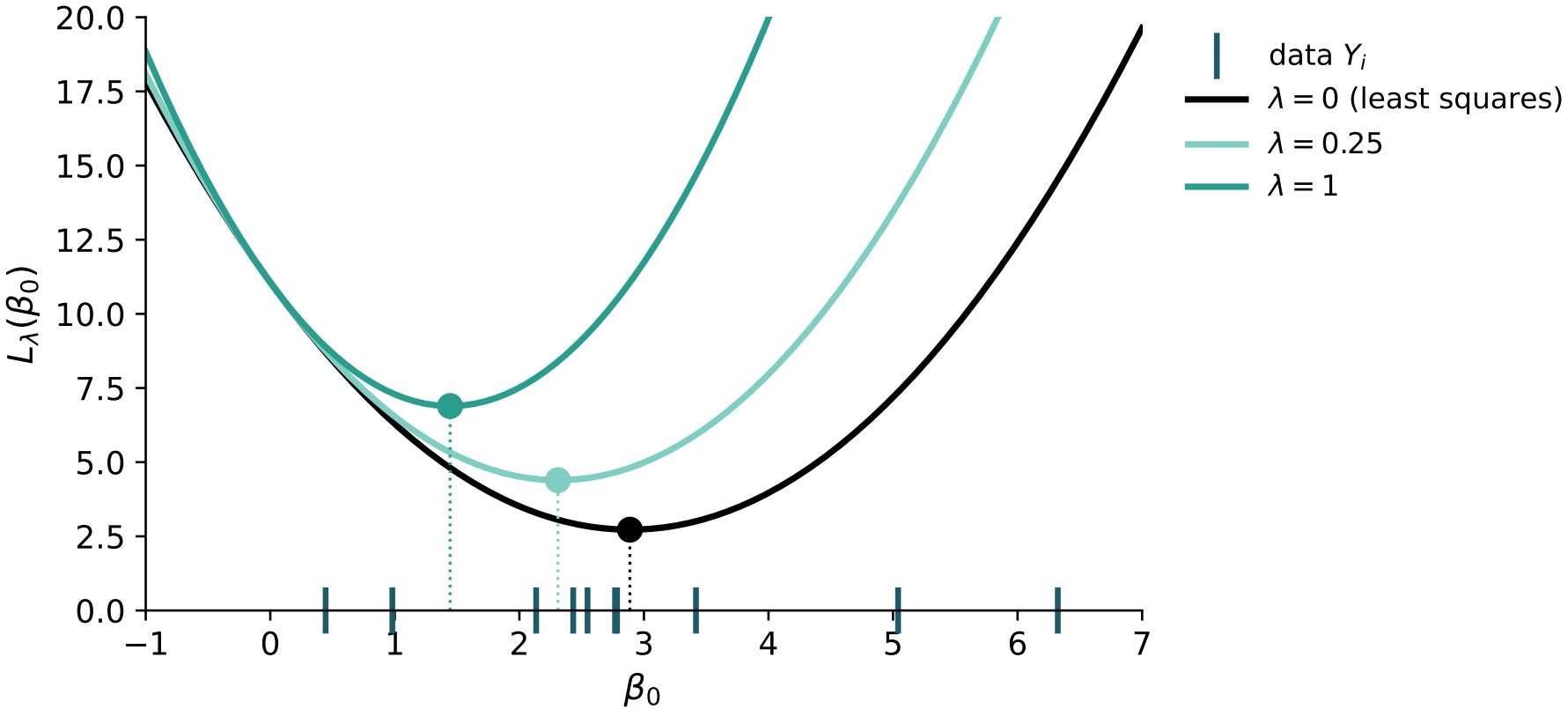
The loss for the mean, for several λ



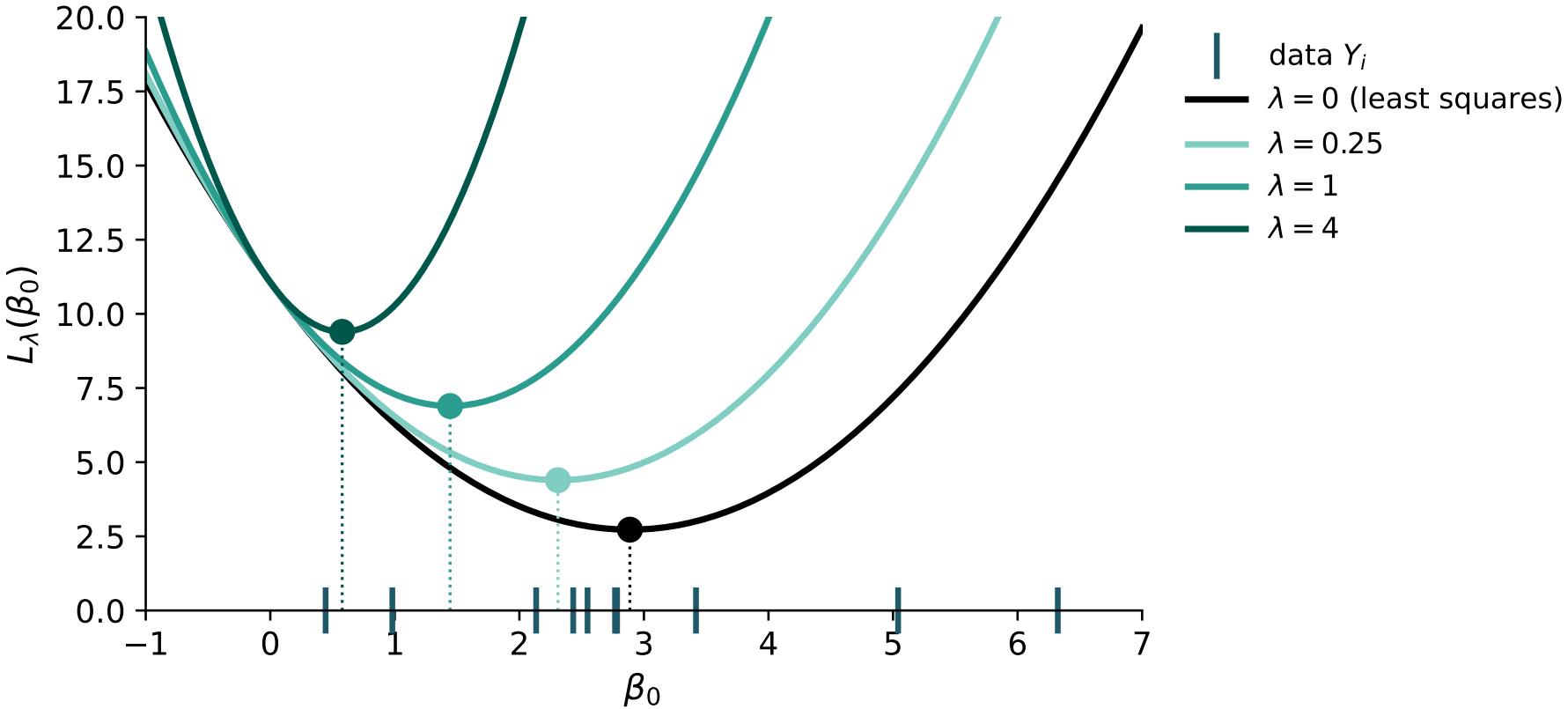
The loss for the mean, for several λ



The loss for the mean, for several λ



The loss for the mean, for several λ



Regularization for a linear model

Model $Y_i = \beta_1 X_i + \epsilon_i$, with $E[X] = 0$ and no intercept. Write $\hat{\sigma}_X^2 = \frac{1}{N}\sum_i X_i^2$ and $\hat{\beta}_1 = \sum_i X_i Y_i / \sum_i X_i^2$ for the least-squares estimate.

$$L_\lambda(\beta_1) = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \beta_1 X_i)^2 + \lambda\beta_1^2$$

Regularization for a linear model

Model $Y_i = \beta_1 X_i + \epsilon_i$, with $E[X] = 0$ and no intercept. Write $\hat{\sigma}_X^2 = \frac{1}{N}\sum_i X_i^2$ and $\hat{\beta}_1 = \sum_i X_i Y_i / \sum_i X_i^2$ for the least-squares estimate.

$$L_\lambda(\beta_1) = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \beta_1 X_i)^2 + \lambda\beta_1^2$$

Using $\frac{1}{N}\sum_i X_i Y_i = \hat{\sigma}_X^2\hat{\beta}_1$:

$$\frac{dL_\lambda}{d\beta_1} = -\frac{2}{N}\sum_{i=1}^{N} X_i(Y_i - \beta_1 X_i) + 2\lambda\beta_1$$

Regularization for a linear model

Model $Y_i = \beta_1 X_i + \epsilon_i$, with $E[X] = 0$ and no intercept. Write $\hat{\sigma}_X^2 = \frac{1}{N}\sum_i X_i^2$ and $\hat{\beta}_1 = \sum_i X_i Y_i / \sum_i X_i^2$ for the least-squares estimate.

$$L_\lambda(\beta_1) = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \beta_1 X_i)^2 + \lambda\beta_1^2$$

Using $\frac{1}{N}\sum_i X_i Y_i = \hat{\sigma}_X^2\hat{\beta}_1$:

$$\begin{aligned}\frac{dL_\lambda}{d\beta_1} &= -\frac{2}{N}\sum_{i=1}^{N} X_i(Y_i - \beta_1 X_i) + 2\lambda\beta_1 \\ &= -2\hat{\sigma}_X^2\hat{\beta}_1 + 2\hat{\sigma}_X^2\beta_1 + 2\lambda\beta_1 = 0\end{aligned}$$

Regularization for a linear model

Model $Y_i = \beta_1 X_i + \epsilon_i$, with $E[X] = 0$ and no intercept. Write $\hat{\sigma}_X^2 = \frac{1}{N}\sum_i X_i^2$ and $\hat{\beta}_1 = \sum_i X_i Y_i / \sum_i X_i^2$ for the least-squares estimate.

$$L_\lambda(\beta_1) = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \beta_1 X_i)^2 + \lambda\beta_1^2$$

Using $\frac{1}{N}\sum_i X_i Y_i = \hat{\sigma}_X^2\hat{\beta}_1$:

$$\begin{aligned}\frac{dL_\lambda}{d\beta_1} &= -\frac{2}{N}\sum_{i=1}^{N} X_i(Y_i - \beta_1 X_i) + 2\lambda\beta_1 \\ &= -2\hat{\sigma}_X^2\hat{\beta}_1 + 2\hat{\sigma}_X^2\beta_1 + 2\lambda\beta_1 = 0\end{aligned}$$

So the minimizer is

$$\hat{\beta}_1^\lambda = \frac{\hat{\sigma}_X^2}{\hat{\sigma}_X^2 + \lambda}\hat{\beta}_1,$$

the least-squares slope shrunk toward 0. The more spread out the X_i (larger $\hat{\sigma}_X^2$), the less a given λ shrinks it.

Idea of regularization

least squares, degree K

$\hat{\beta}$ minimizes $\quad L(\beta) = \hat{\sigma}^2_{\text{train}} = \dfrac{1}{N_{\text{train}}} \displaystyle\sum_{i=1}^{N_{\text{train}}} \left(\hat{y}(X_i^{\text{train}}, D^{\text{train}}) - Y_i^{\text{train}}\right)^2$

each x^j is either in the model ($j \leq K$) or not

Idea of regularization

least squares, degree K

$\hat{\beta}$ minimizes $\quad L(\beta) = \hat{\sigma}^2_{\text{train}} = \dfrac{1}{N_{\text{train}}} \displaystyle\sum_{i=1}^{N_{\text{train}}} \left(\hat{y}(X_i^{\text{train}}, D^{\text{train}}) - Y_i^{\text{train}}\right)^2$

each x^j is either in the model ($j \leq K$) or not

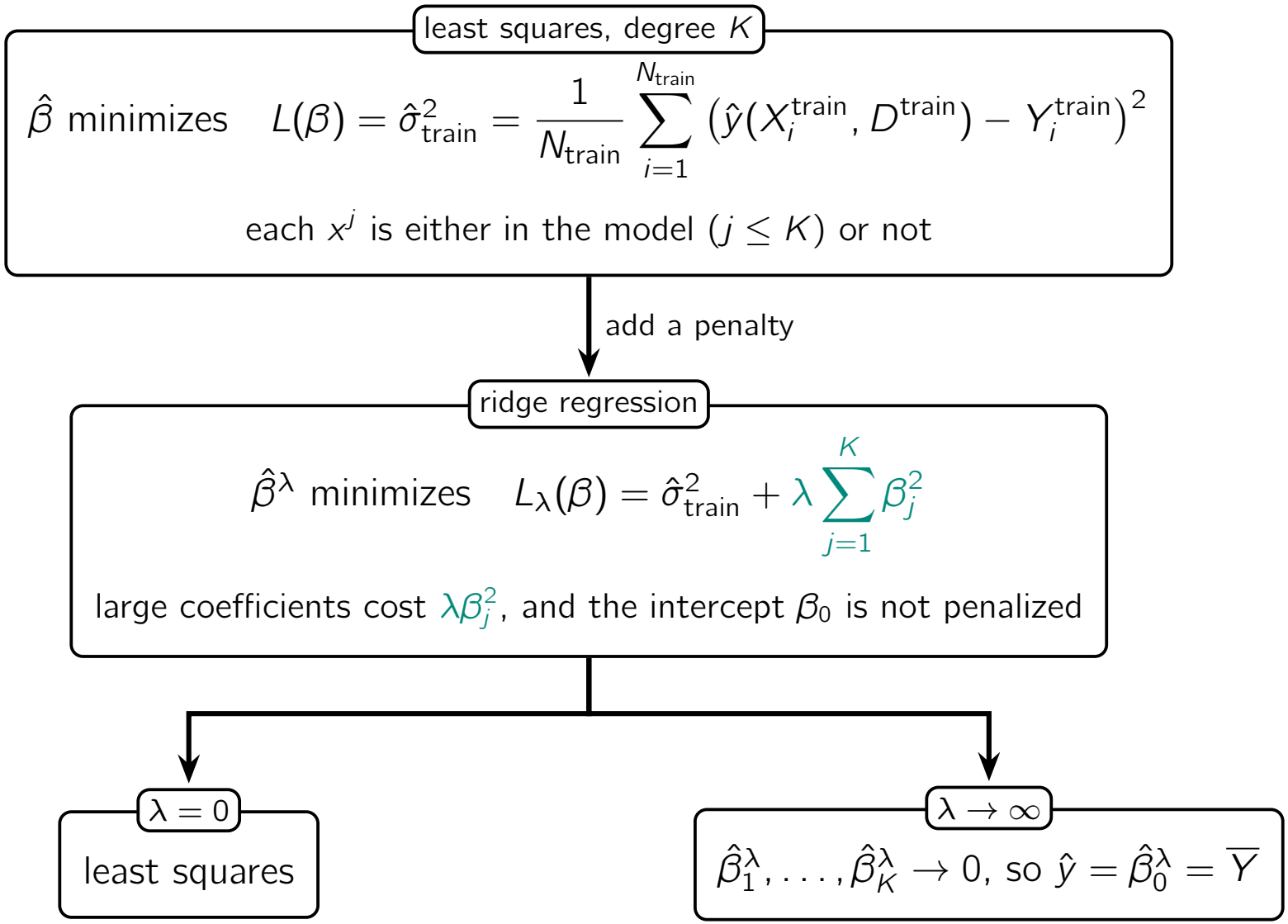
add a penalty

ridge regression

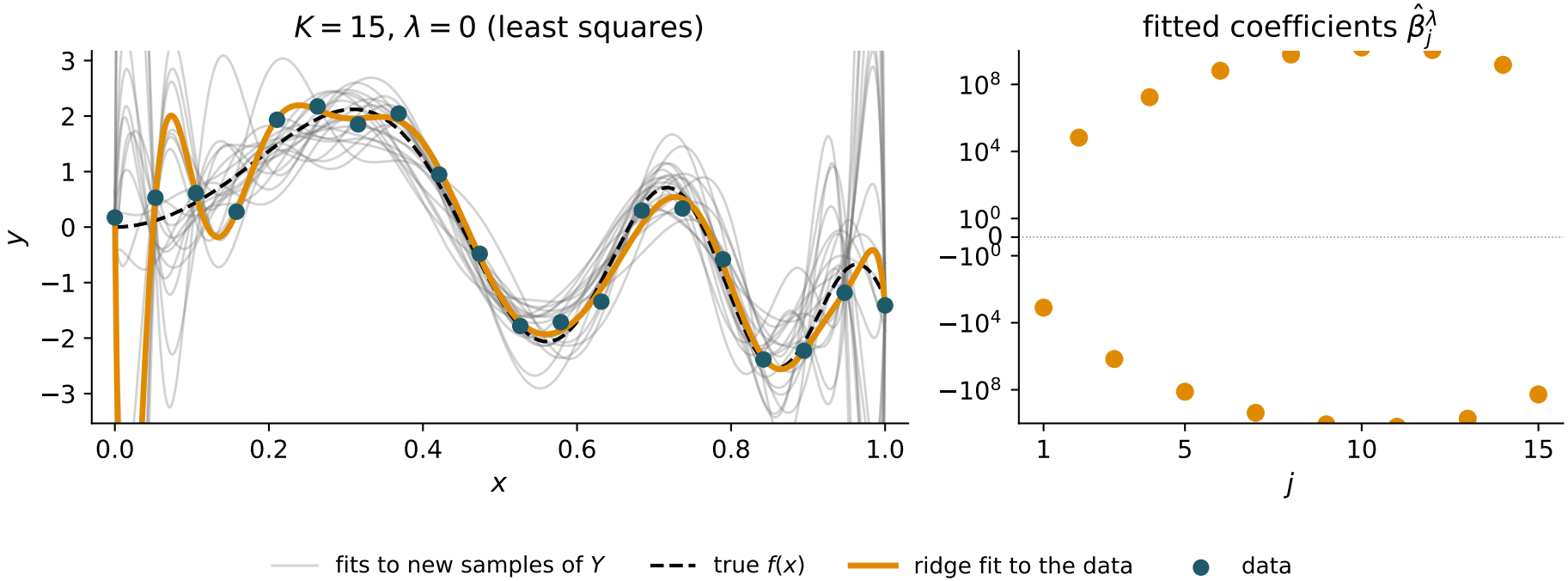
$\hat{\beta}^\lambda$ minimizes $\quad L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \displaystyle\sum_{j=1}^{K} \beta_j^2$

large coefficients cost $\lambda\beta_j^2$, and the intercept β_0 is not penalized

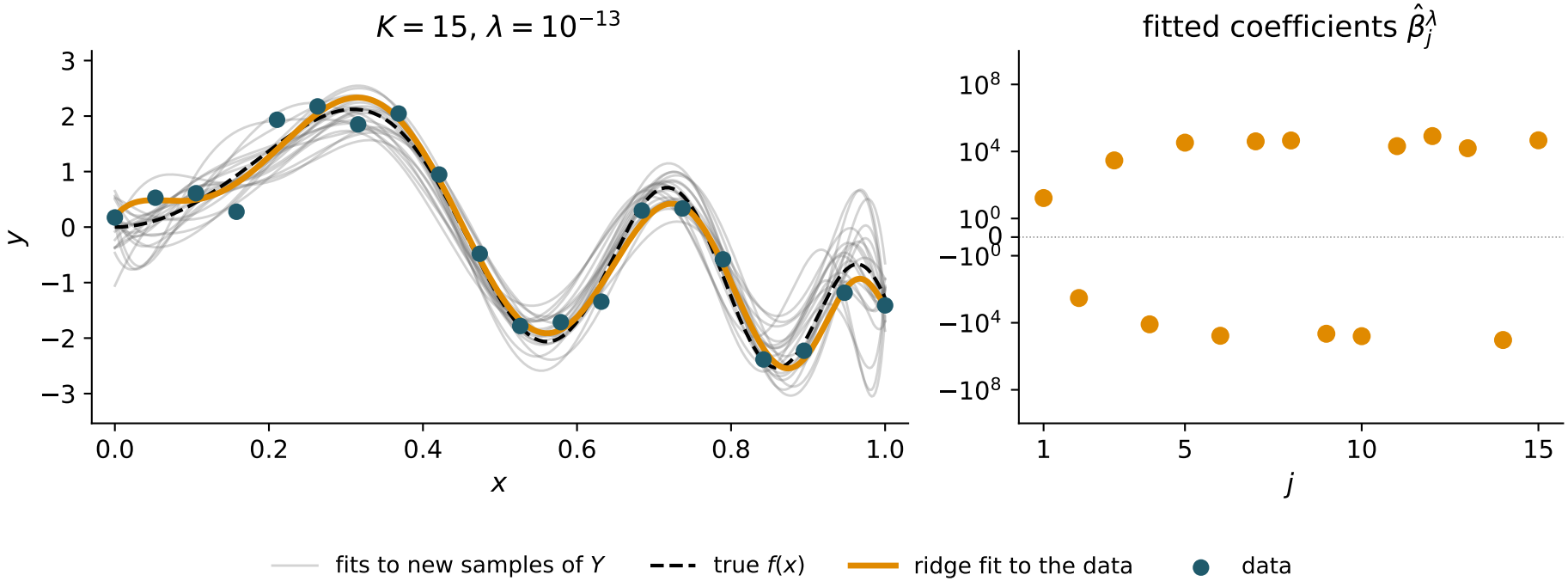
Idea of regularization



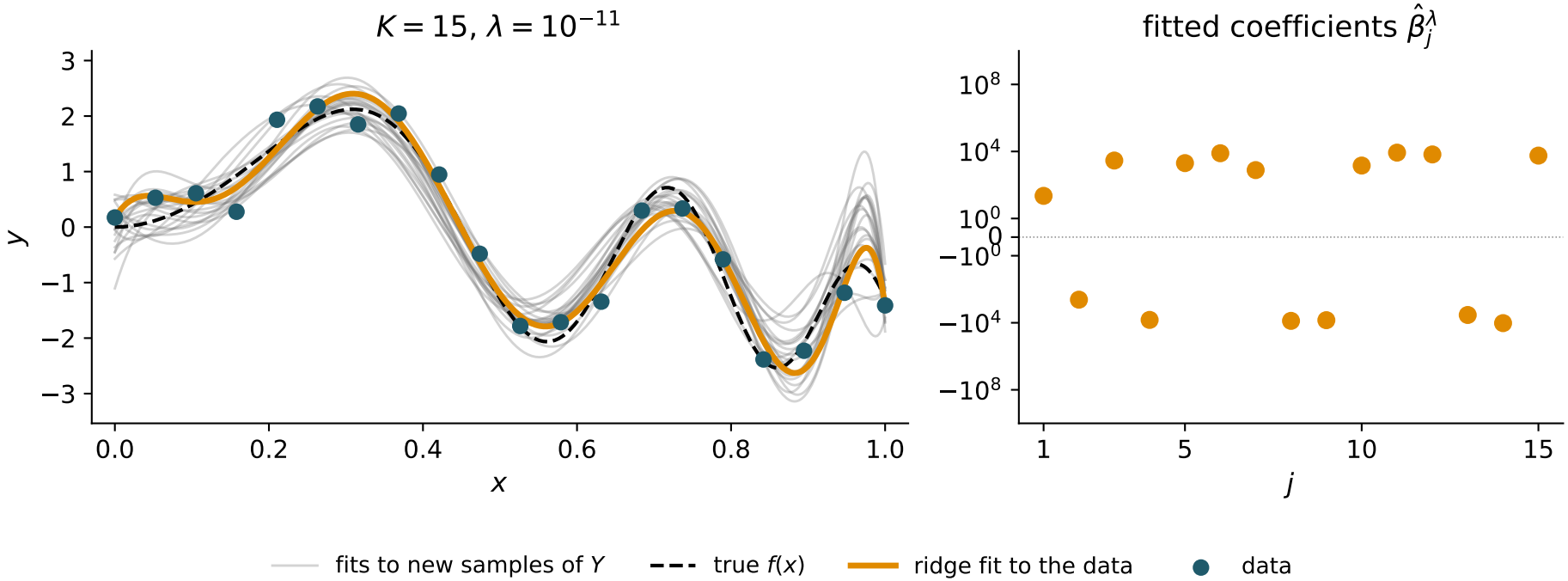
The effect of λ: the 20-point example with $K = 15$



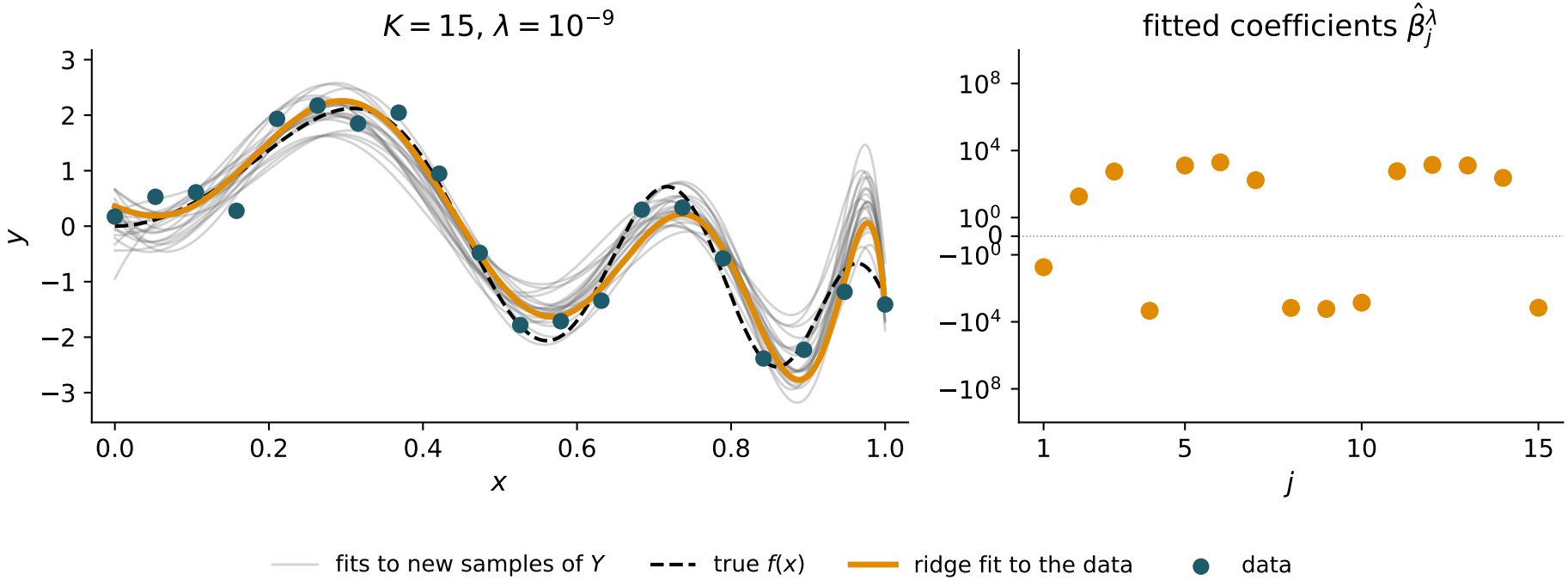
The effect of λ: the 20-point example with $K = 15$



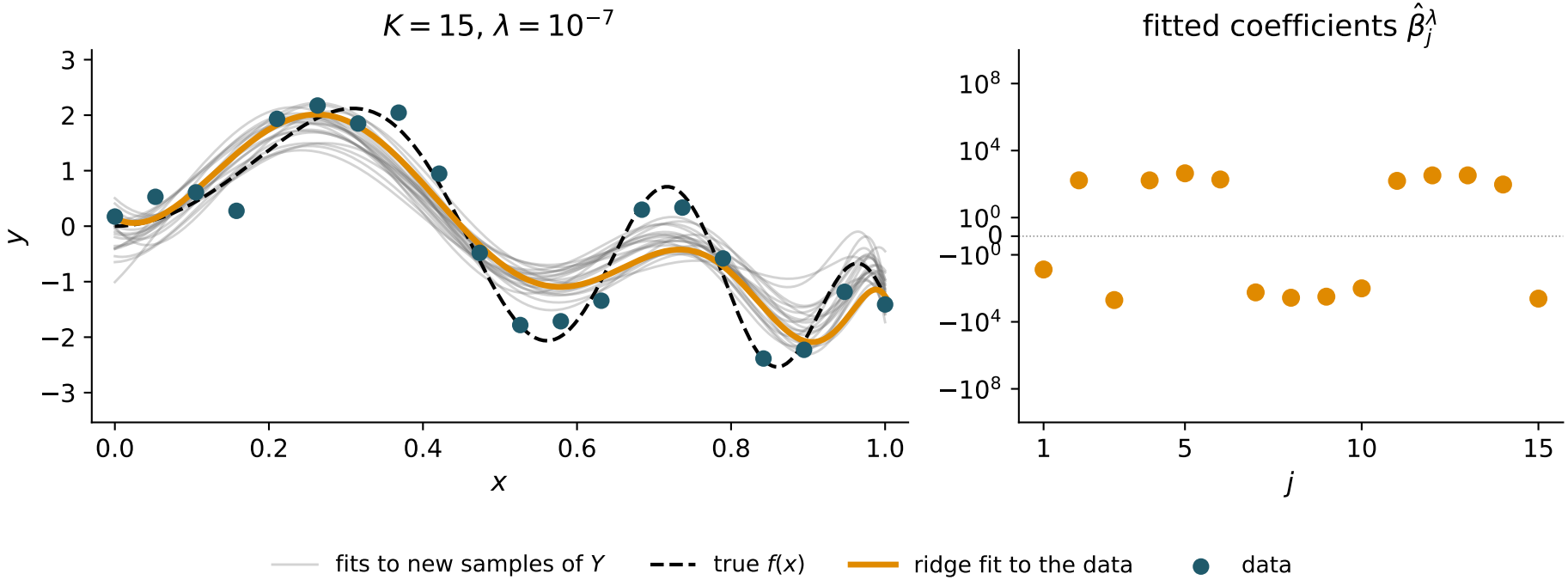
The effect of λ: the 20-point example with $K = 15$



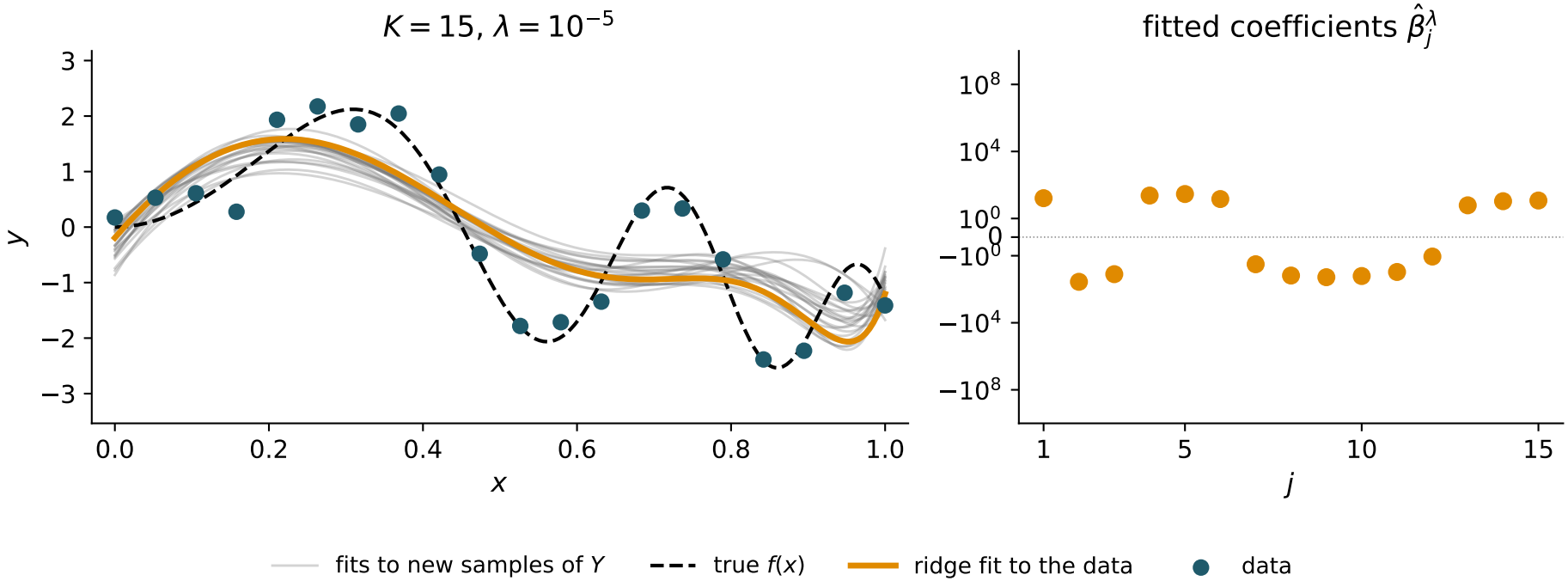
The effect of λ: the 20-point example with $K = 15$



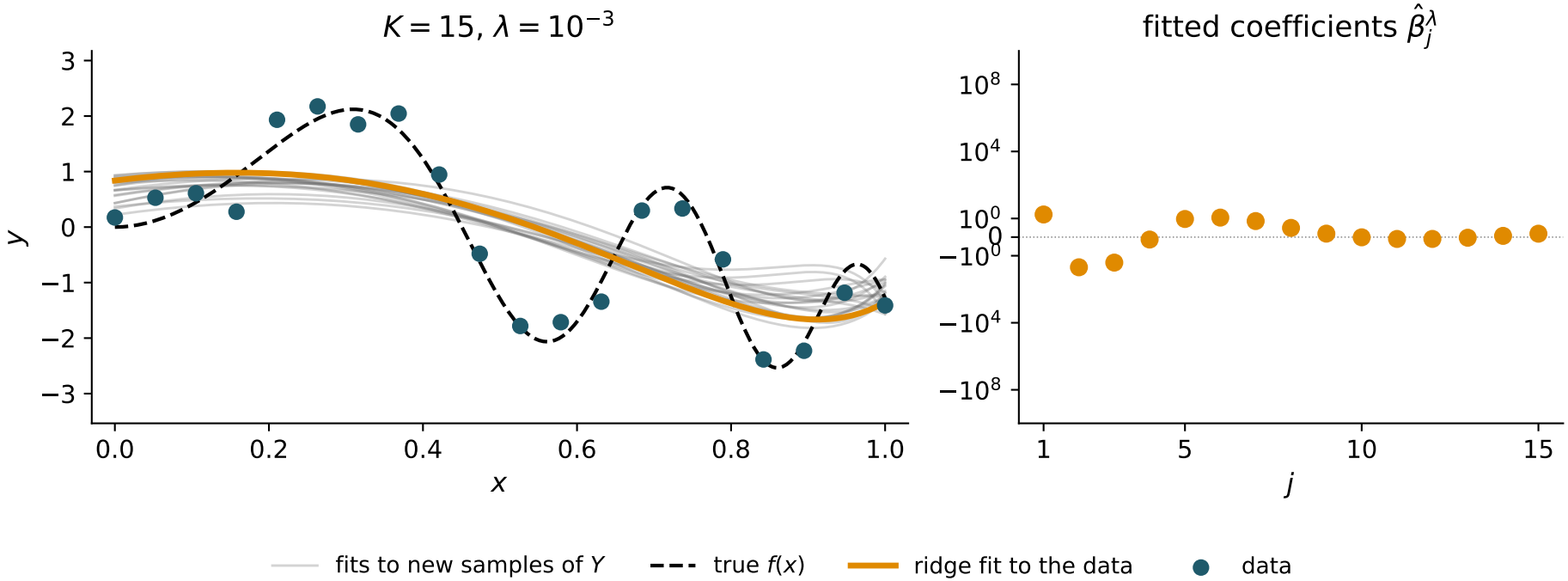
The effect of λ: the 20-point example with $K = 15$



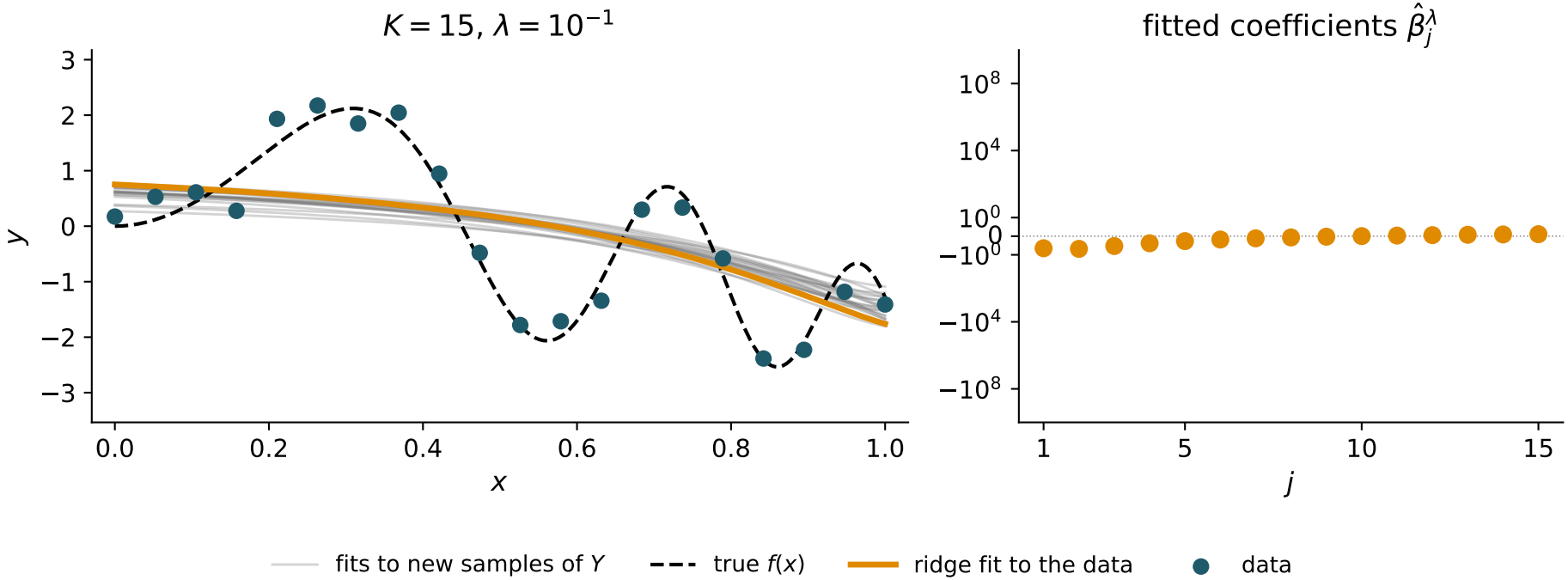
The effect of λ: the 20-point example with $K = 15$



The effect of λ: the 20-point example with $K = 15$

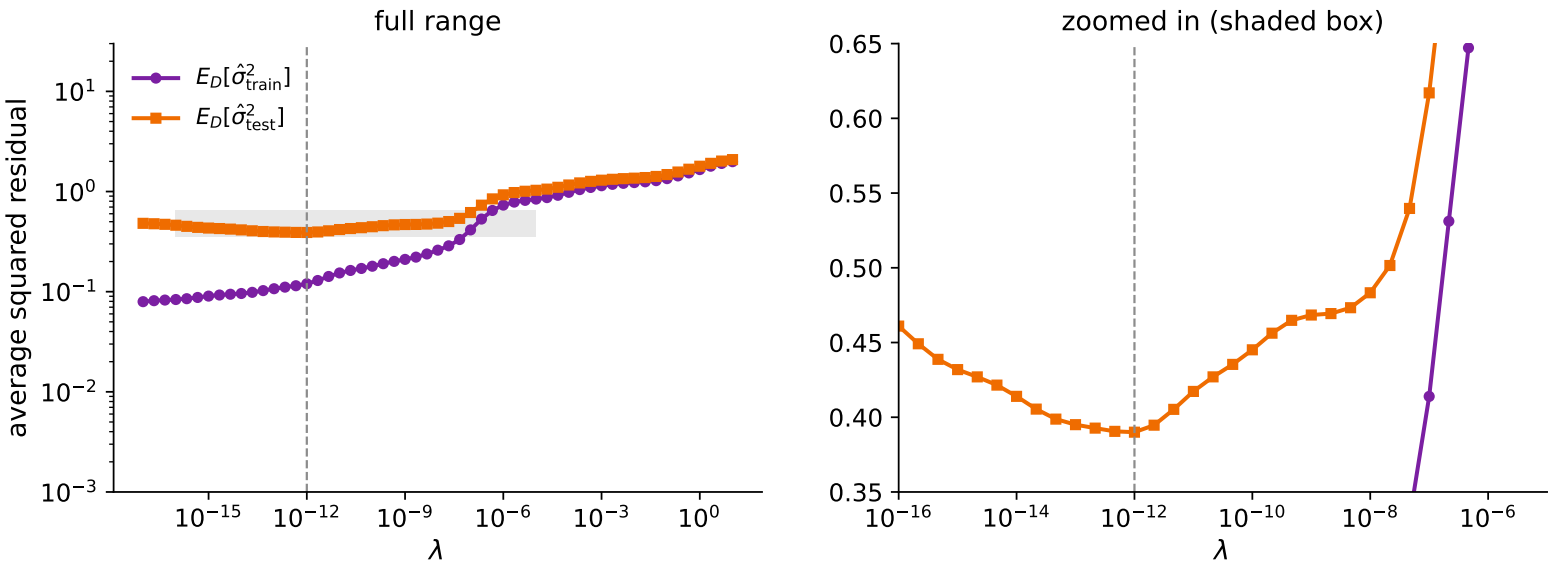


The effect of λ: the 20-point example with $K = 15$



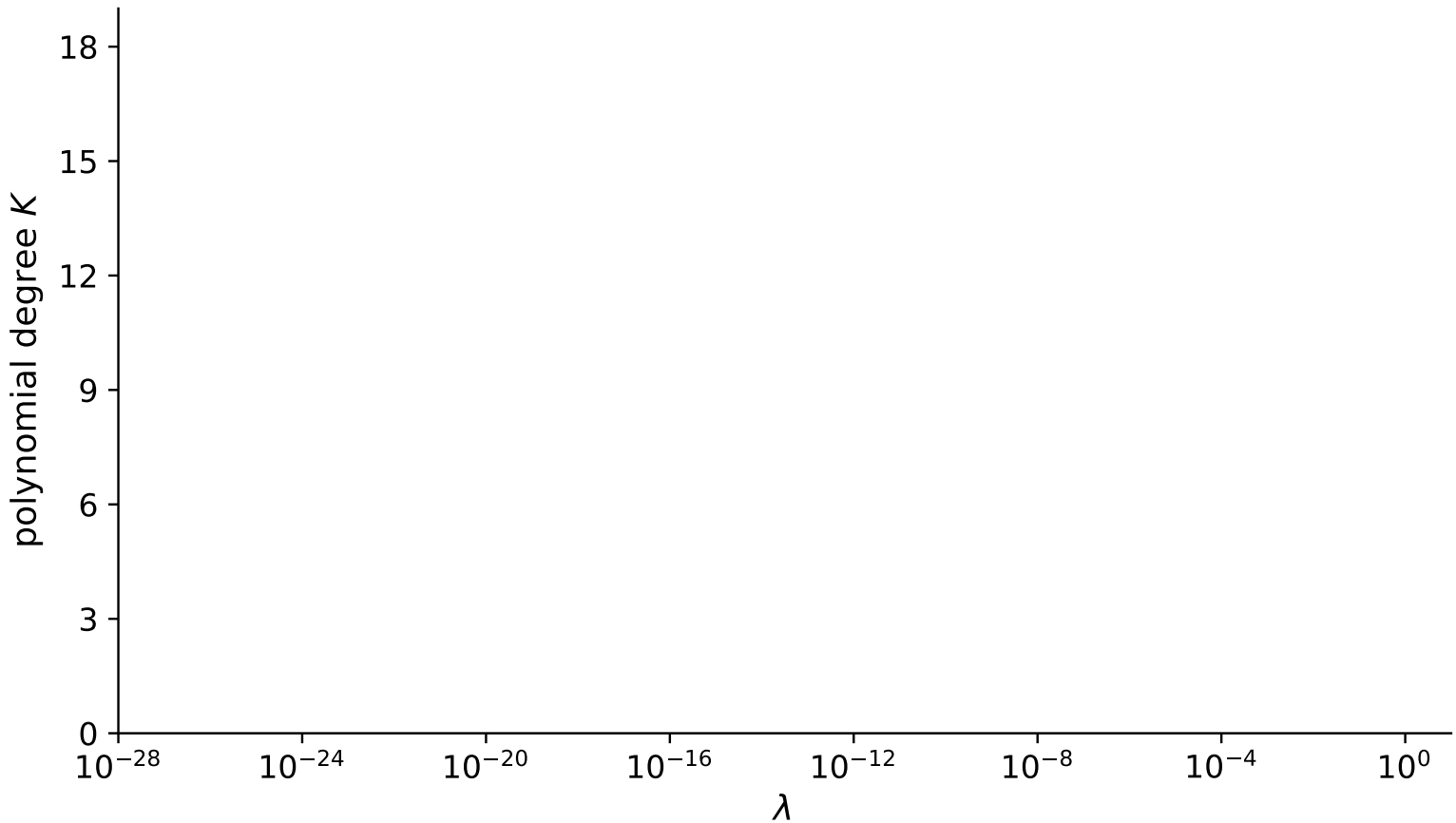
Training and test error vs. λ

Each error is averaged over 400 simulated training sets of the same 20 x values.



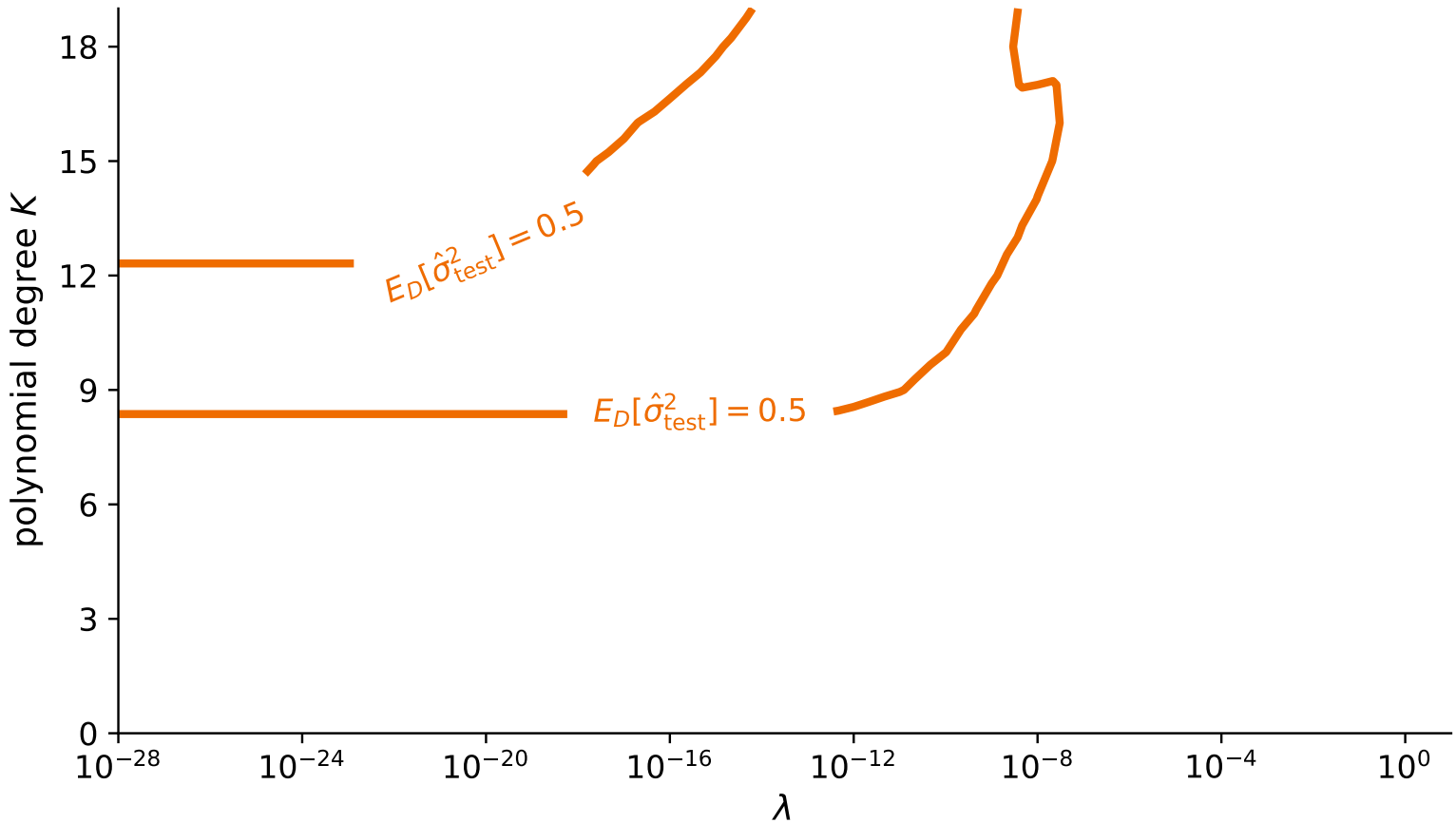
Reading a contour map

Each point is a choice of degree K and penalty λ. A contour line joins the choices that give the same value.



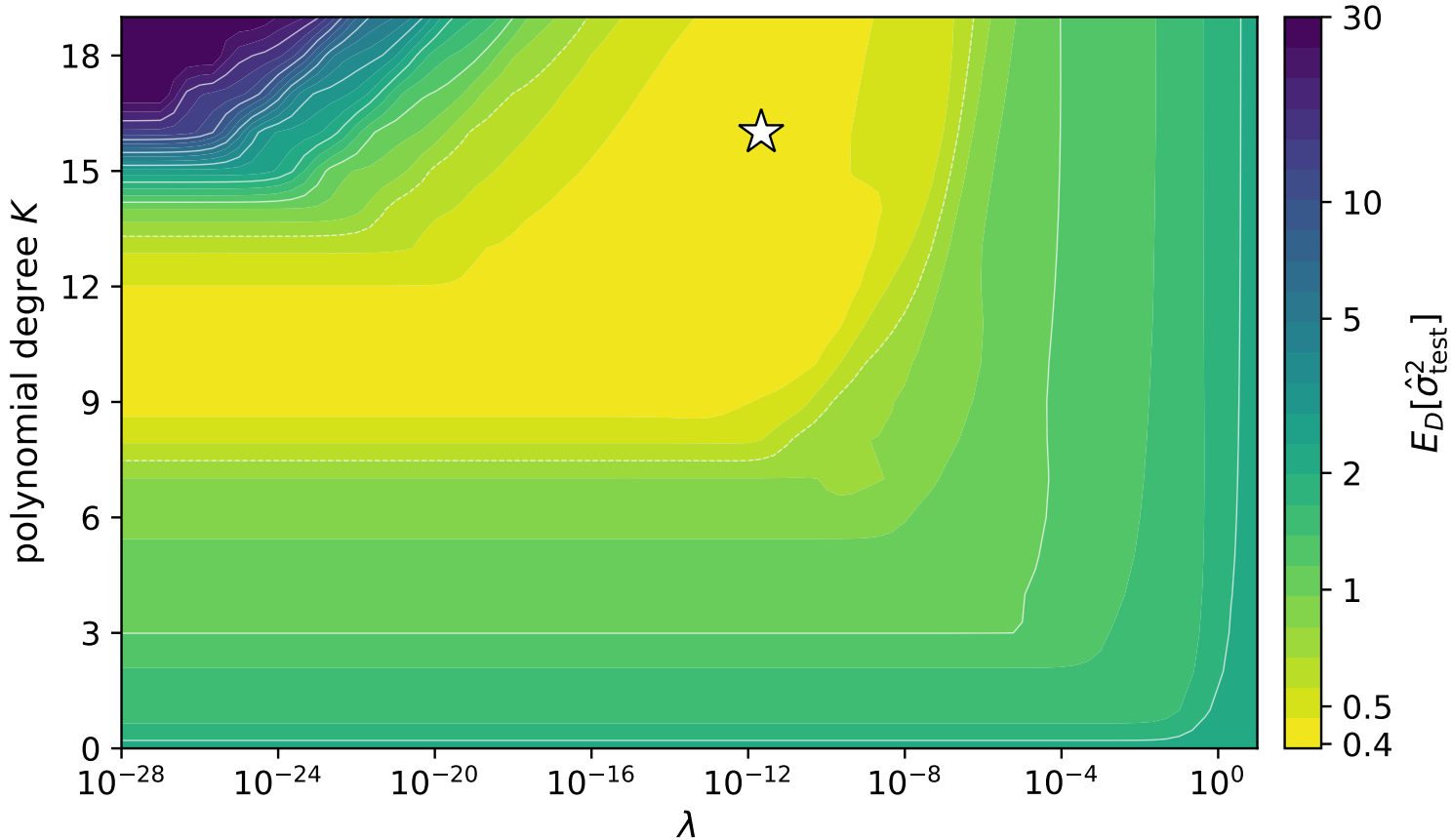
Reading a contour map

Each point is a choice of degree K and penalty λ. A contour line joins the choices that give the same value.

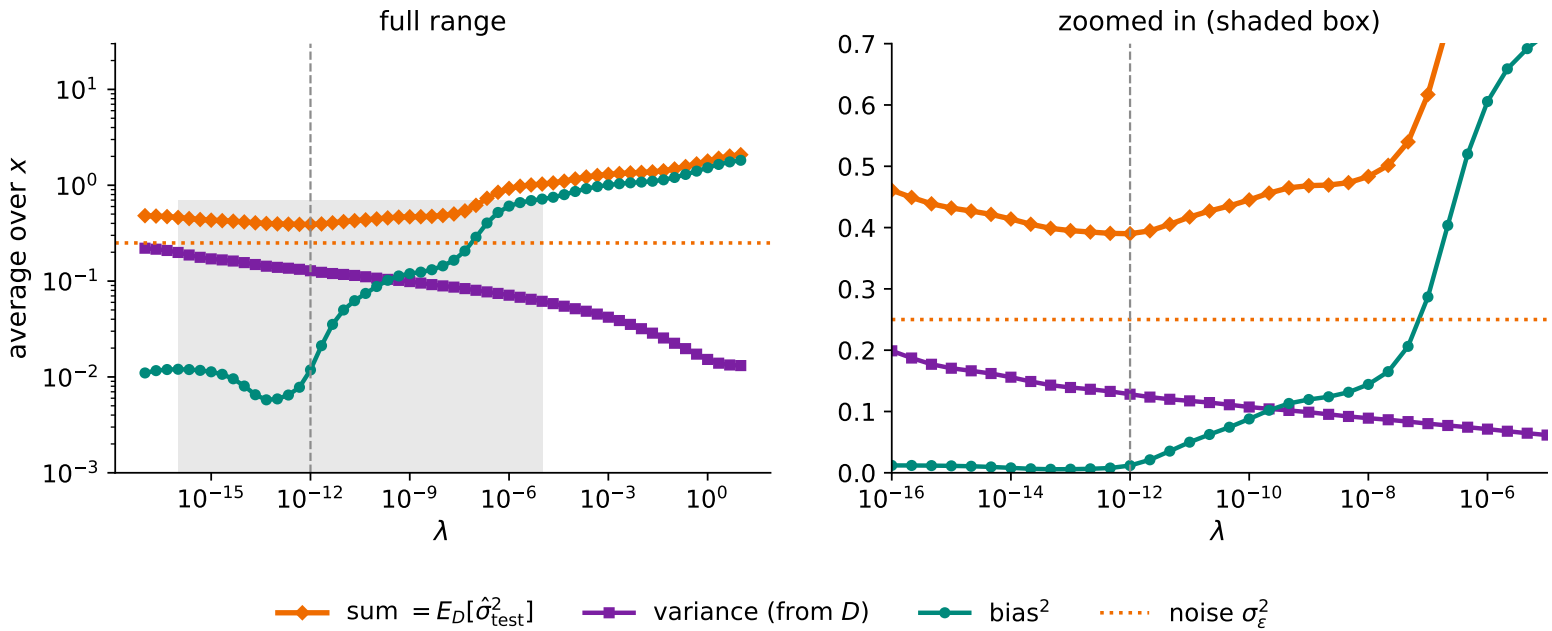


Test error over K and λ

$E_D[\hat{\sigma}^2_{\text{test}}]$ for every degree K and penalty λ, computed exactly. The star is the smallest value.



Bias and variance vs. λ



The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

training set

$N_{\text{train}} = 4000$ variants

fit with ridge regression

minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

every coefficient except β_0 gets the same penalty

The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

training set

$N_{\text{train}} = 4000$ variants

fit with ridge regression

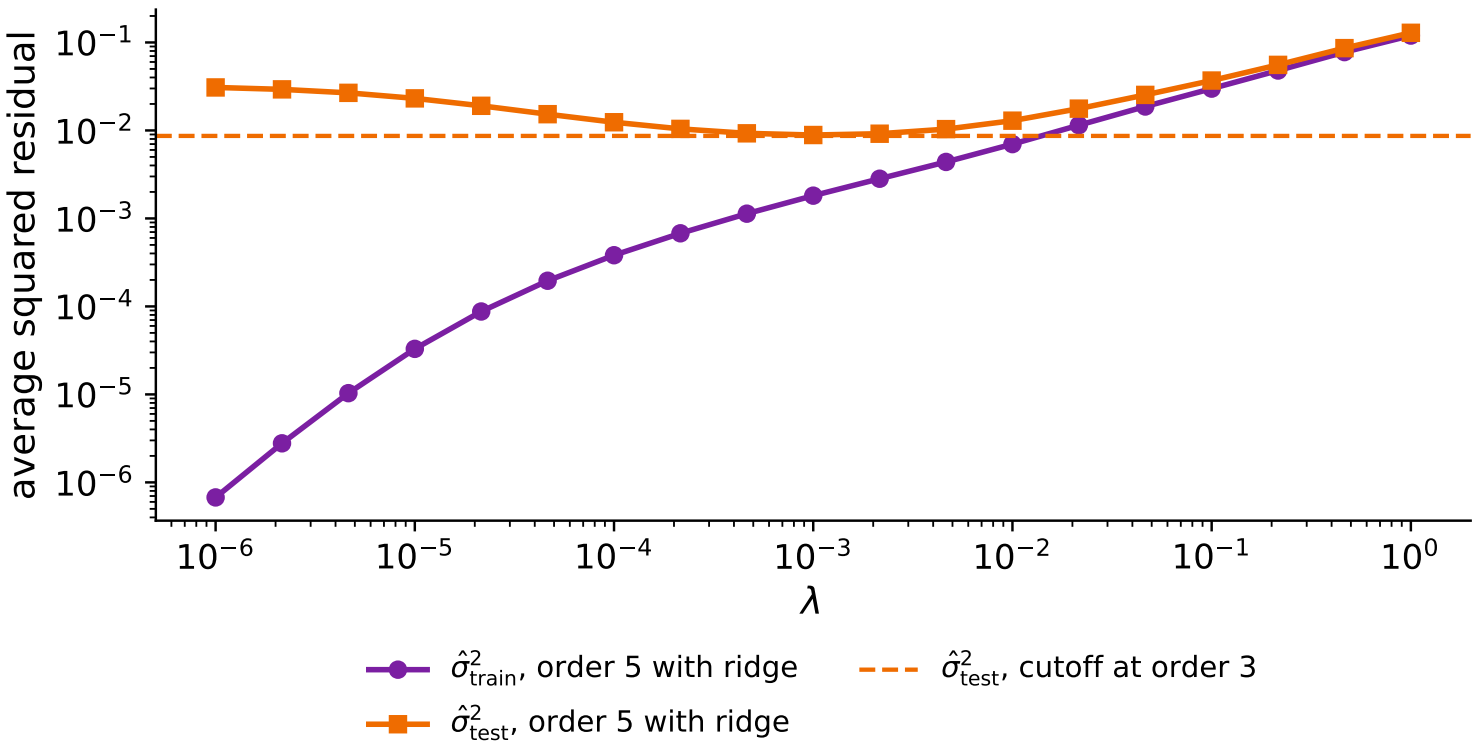
minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

every coefficient except β_0 gets the same penalty

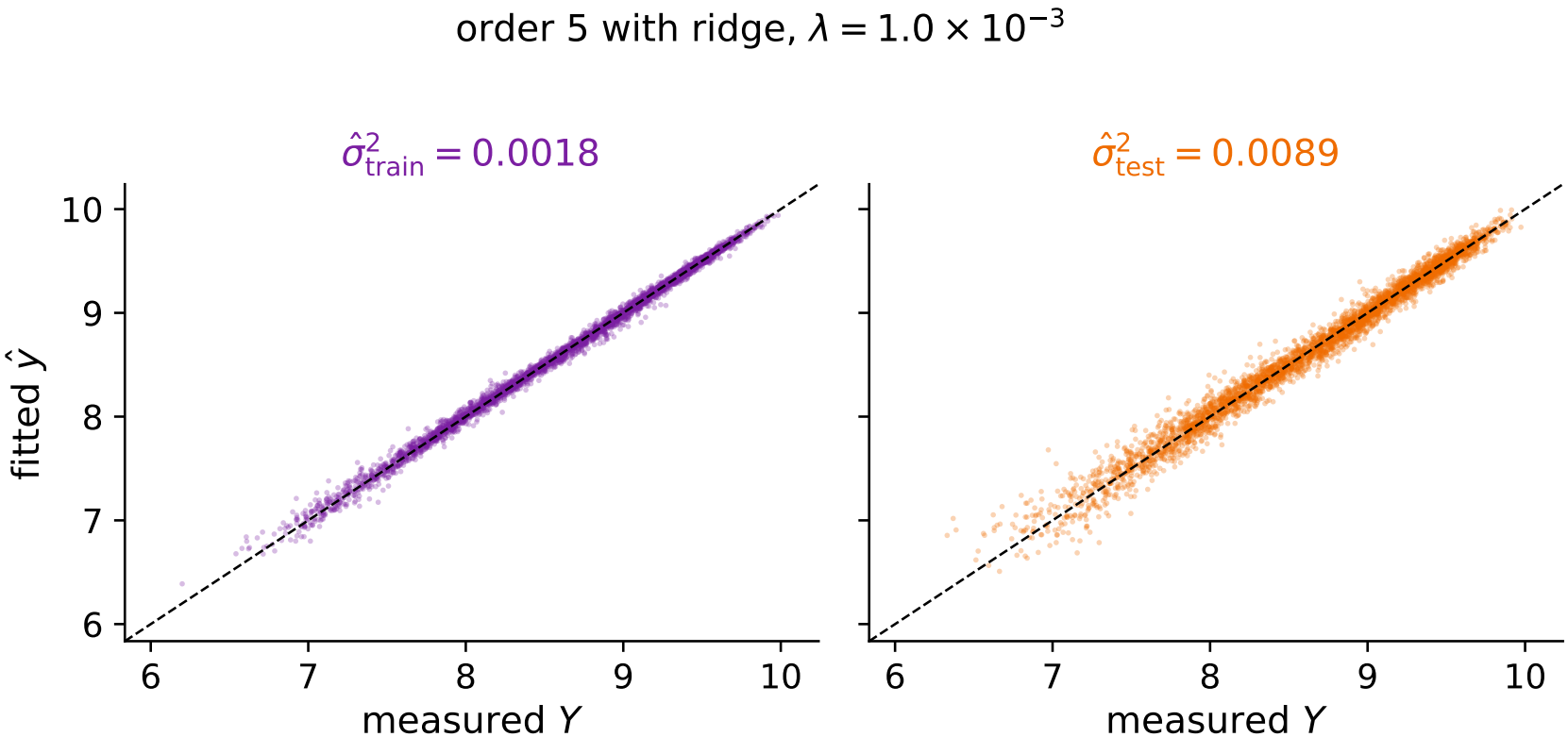
4944 coefficients but only 4000 training variants, so least squares fits the training data exactly. With $\lambda > 0$ the fit is unique and no longer interpolates.

Regularization for the spike protein

All interactions up to order 5 (4944 coefficients), fit with ridge to the same 4000 training variants as on 11/6 and scored on the same 4000 test variants. The dashed line is least squares with the cutoff at order 3.



Fitted vs. measured binding with ridge



Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$

$k, l = 0, \ldots, 159$: one feature for each pixel

probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

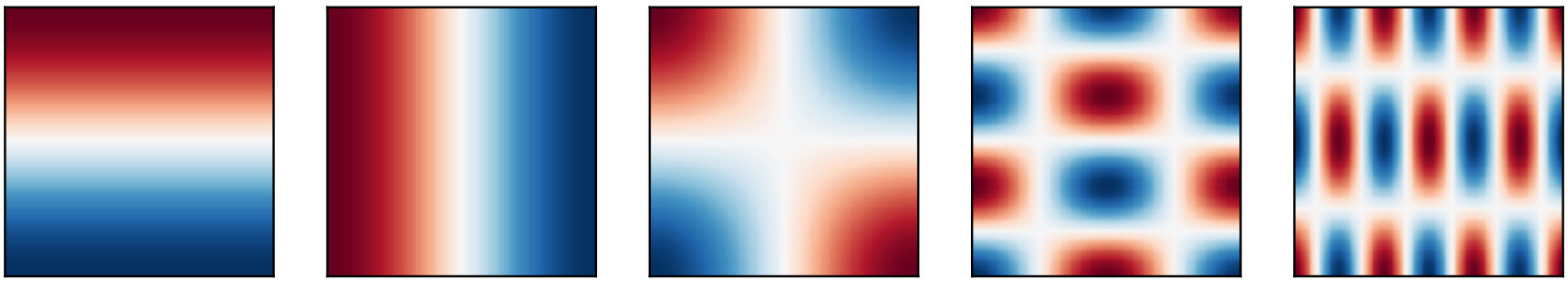
regularization

$$L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \sum_{k,l} (k^2 + l^2)\, \beta^2_{k,l}$$

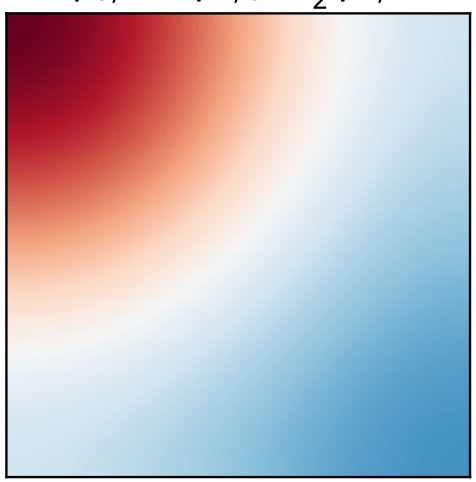
high frequencies cost more

The Fourier features

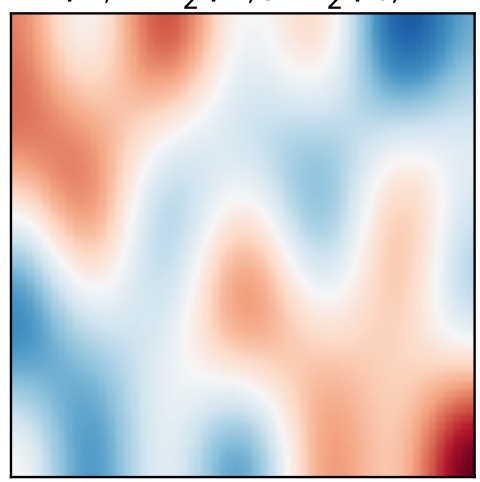
$\phi_{0,1}$ $\phi_{1,0}$ $\phi_{1,1}$ $\phi_{2,3}$ $\phi_{6,2}$



$\phi_{0,1} + \phi_{1,0} + \frac{1}{2}\phi_{1,1}$

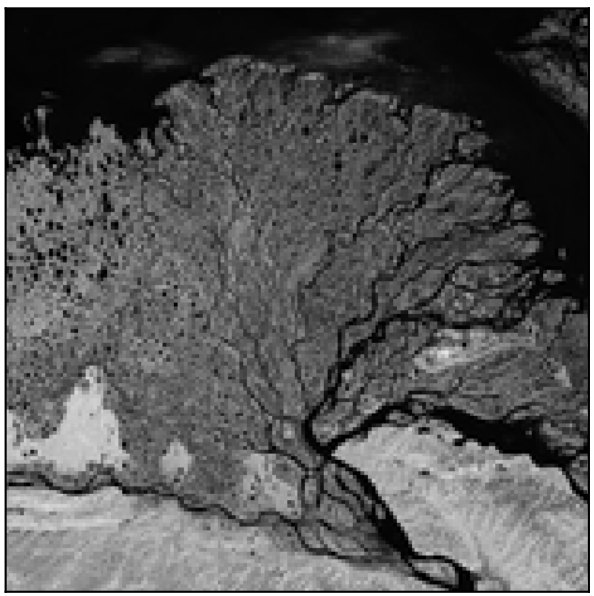


$\phi_{1,1} - \frac{1}{2}\phi_{2,3} + \frac{1}{2}\phi_{6,2}$

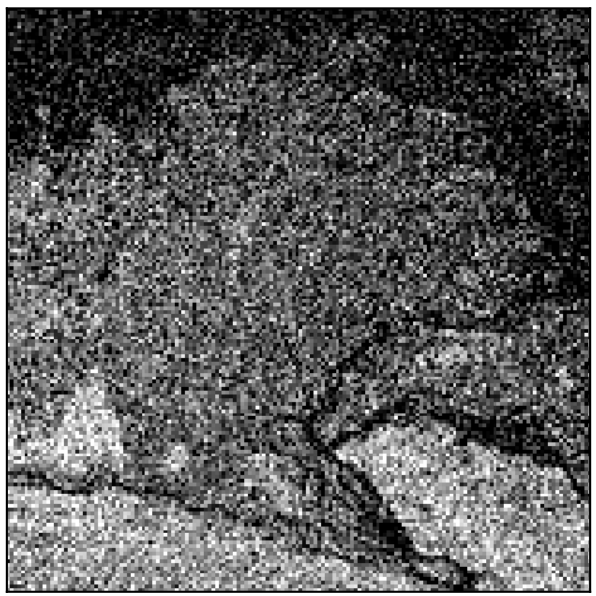


Smoothing a noisy image

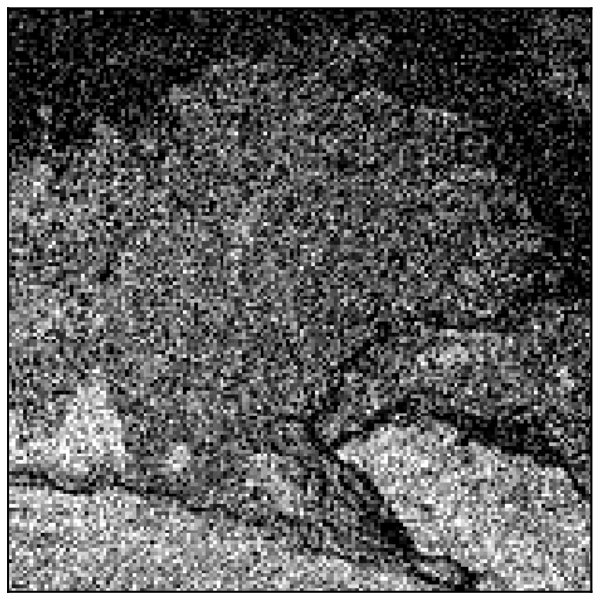
clean image (unknown)



noisy image (data)



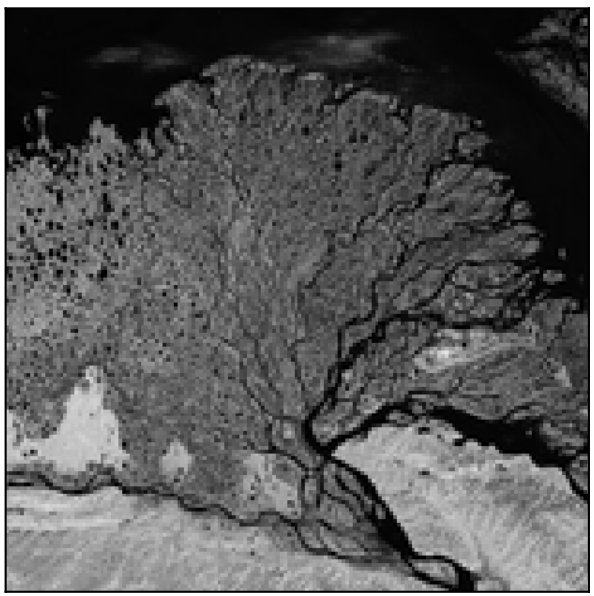
fit, $\lambda = 0$ (least squares)



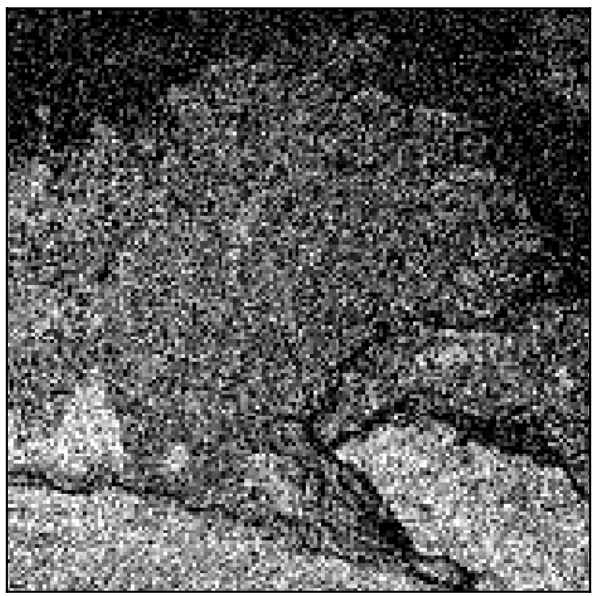
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

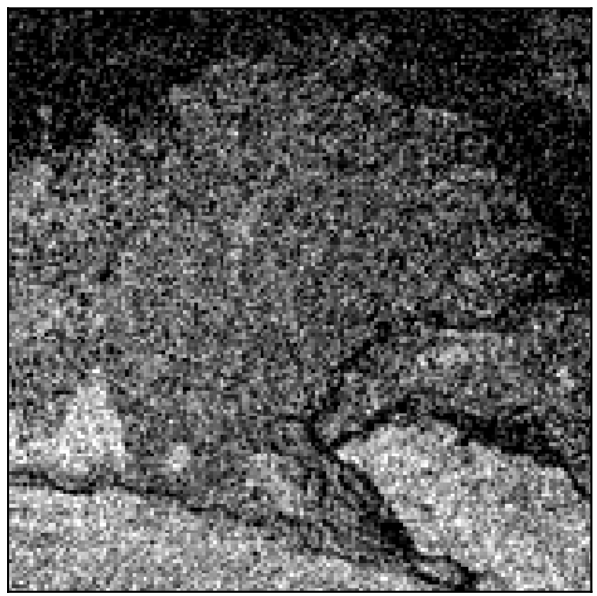
clean image (unknown)



noisy image (data)



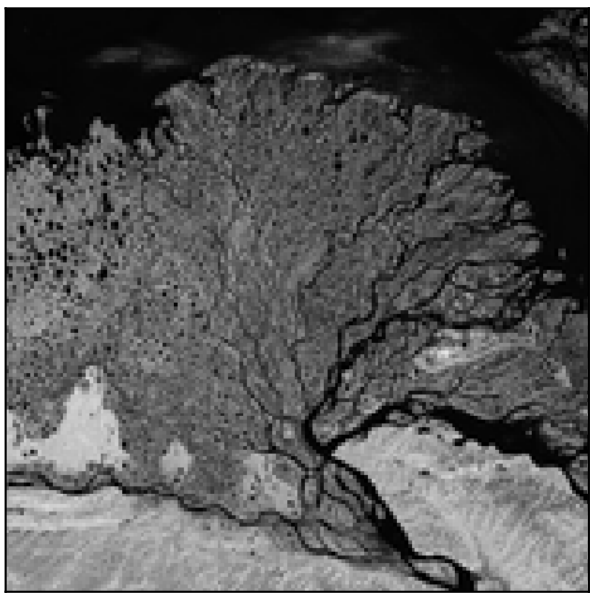
fit, $\lambda = 10^{-9}$



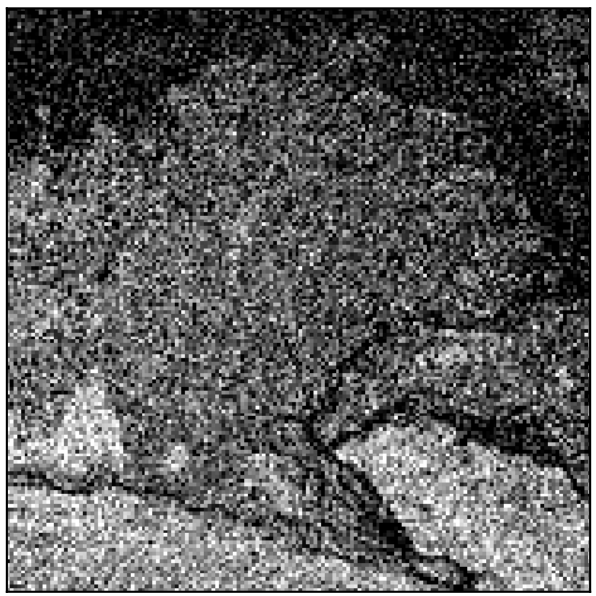
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

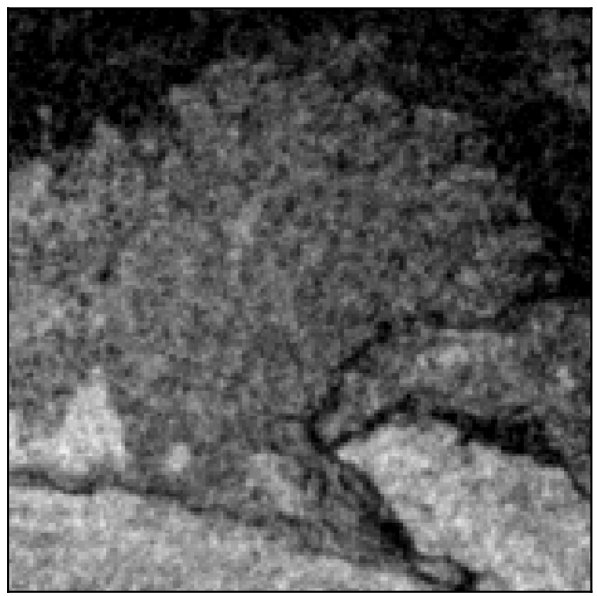
clean image (unknown)



noisy image (data)



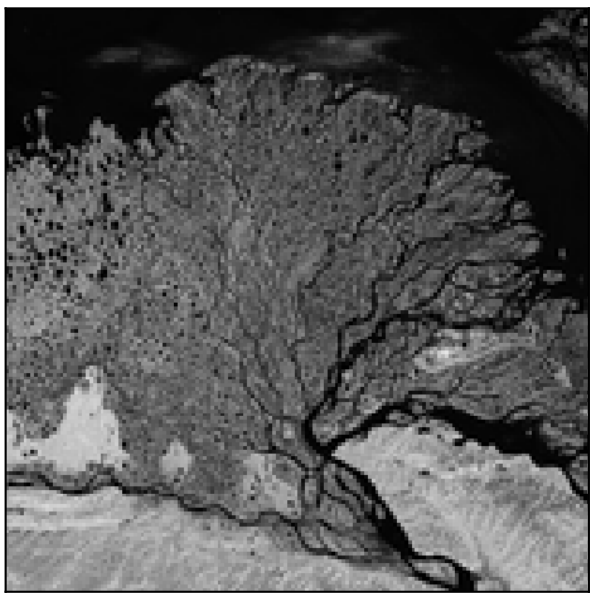
fit, $\lambda = 10^{-8}$



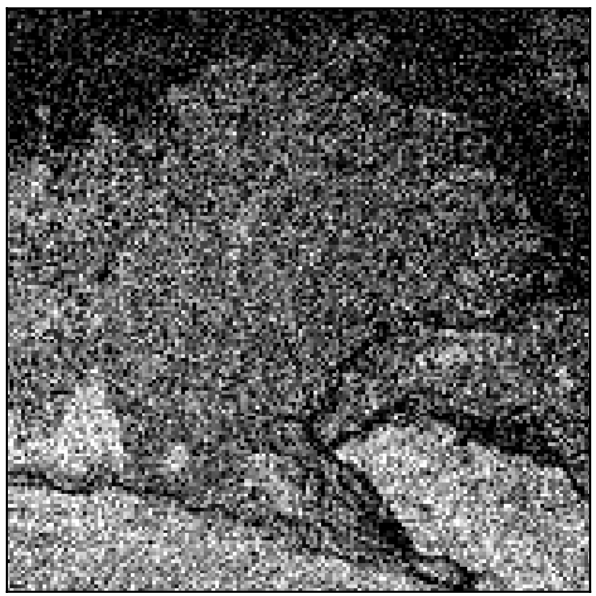
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

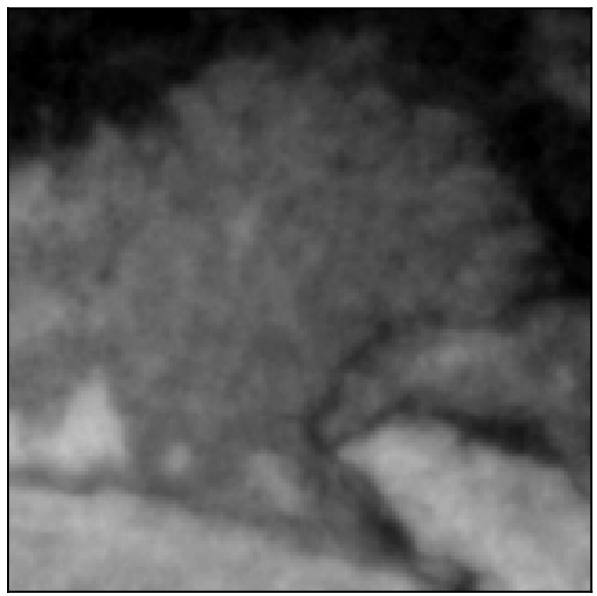
clean image (unknown)



noisy image (data)



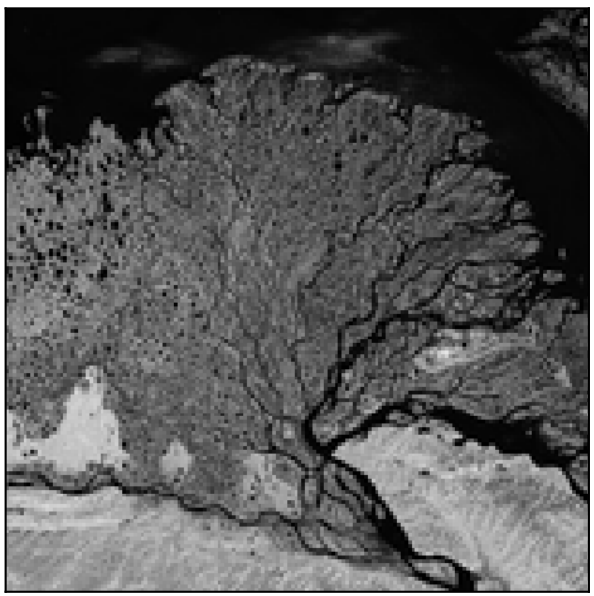
fit, $\lambda = 10^{-7}$



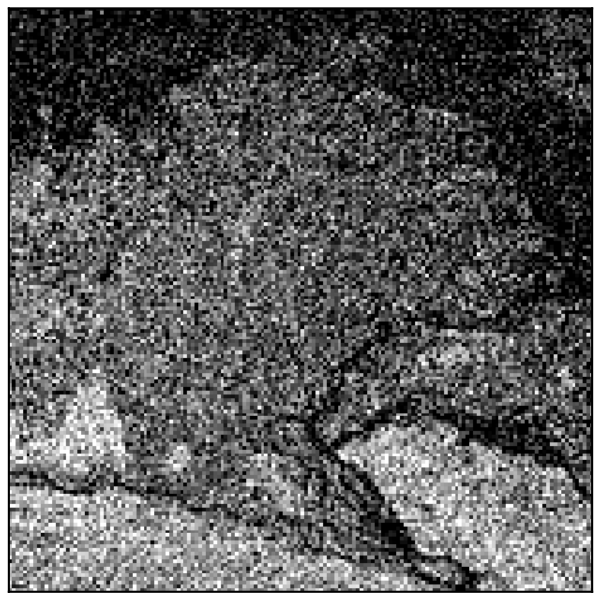
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

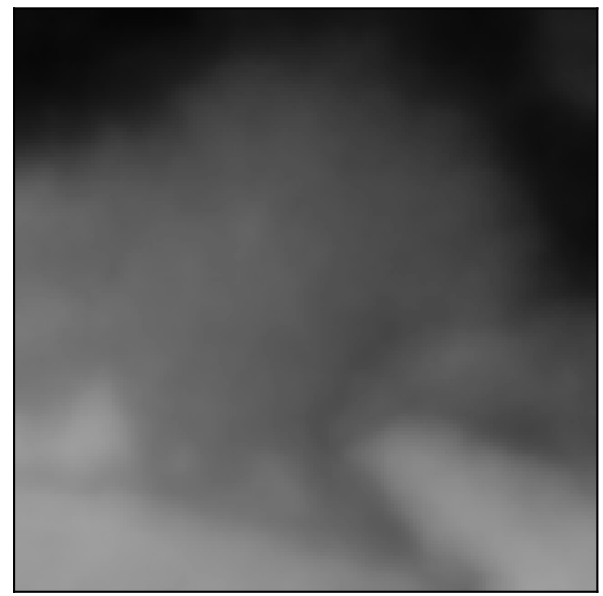
clean image (unknown)



noisy image (data)



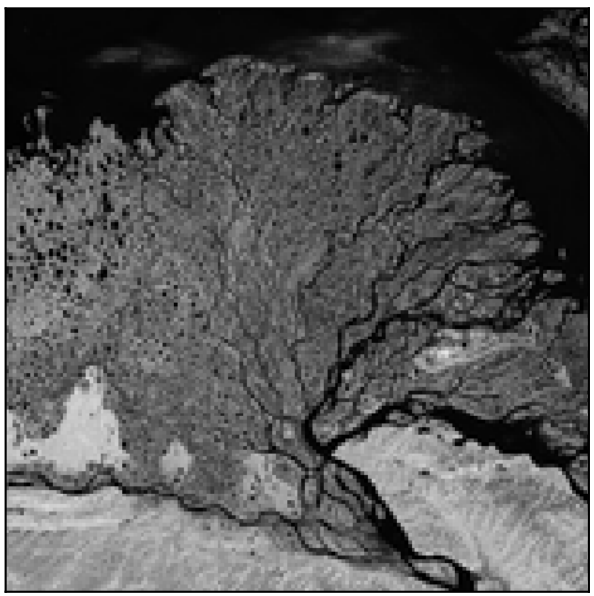
fit, $\lambda = 10^{-6}$



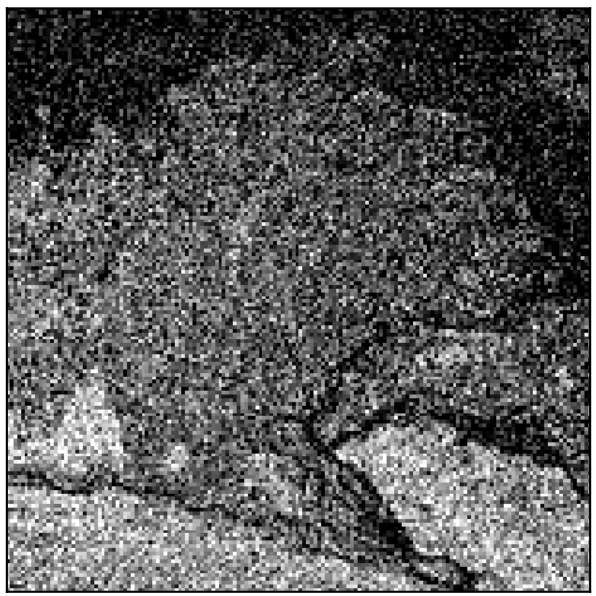
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

clean image (unknown)



noisy image (data)



fit, $\lambda = 10^{-5}$



Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Choosing λ for the image

